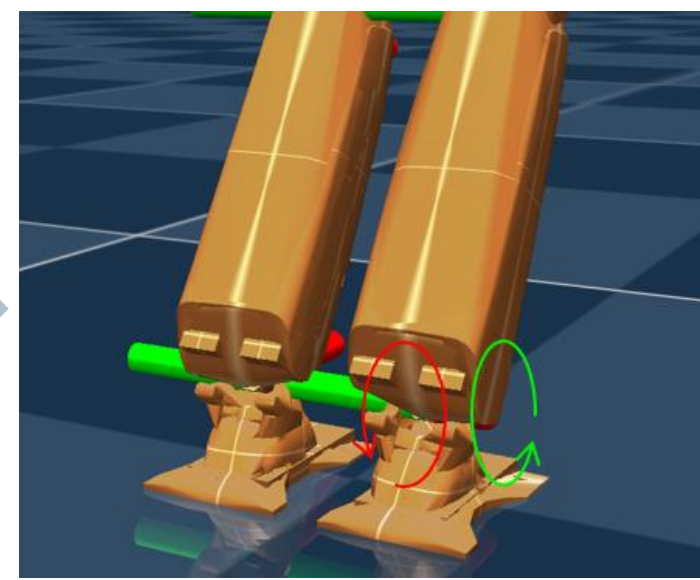
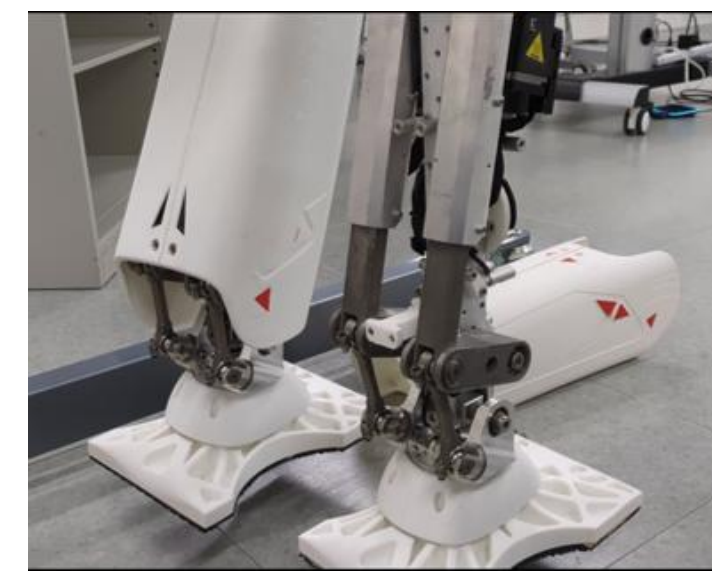


Motion-Dependent Effects of Leg Linkage Modeling on Sim-to-Real Accuracy in Humanoid Robots

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Background

■ Leg linkage mechanism representations for reinforcement learning



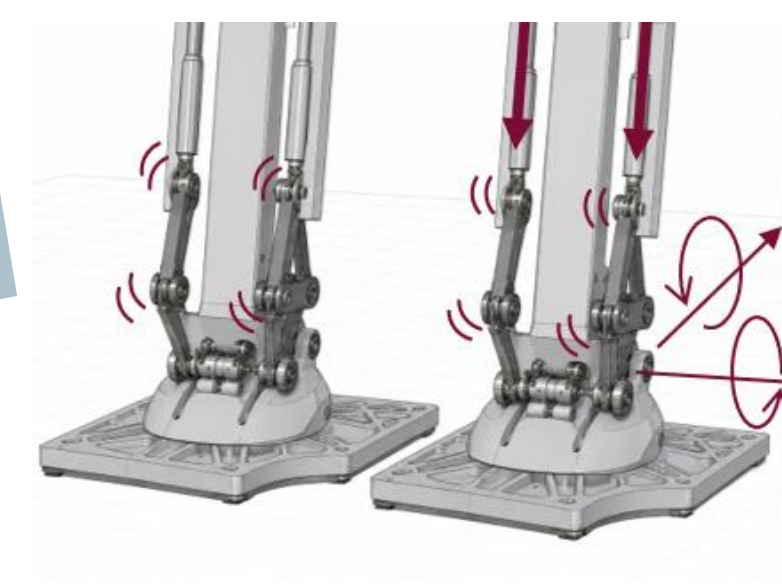
Serial-chain Approximation [1]

- Separate motion learning from linkage dynamics
- Train in the robot's joint space
- Easy to implement and computationally efficient

Closed-chain Linkage Simulation [2]

- Directly simulate the closed linkage mechanism
- Train in the robot's actuator space
- Higher computational cost and implementation complexity, especially when modeling all sub links

Linkage mechanism



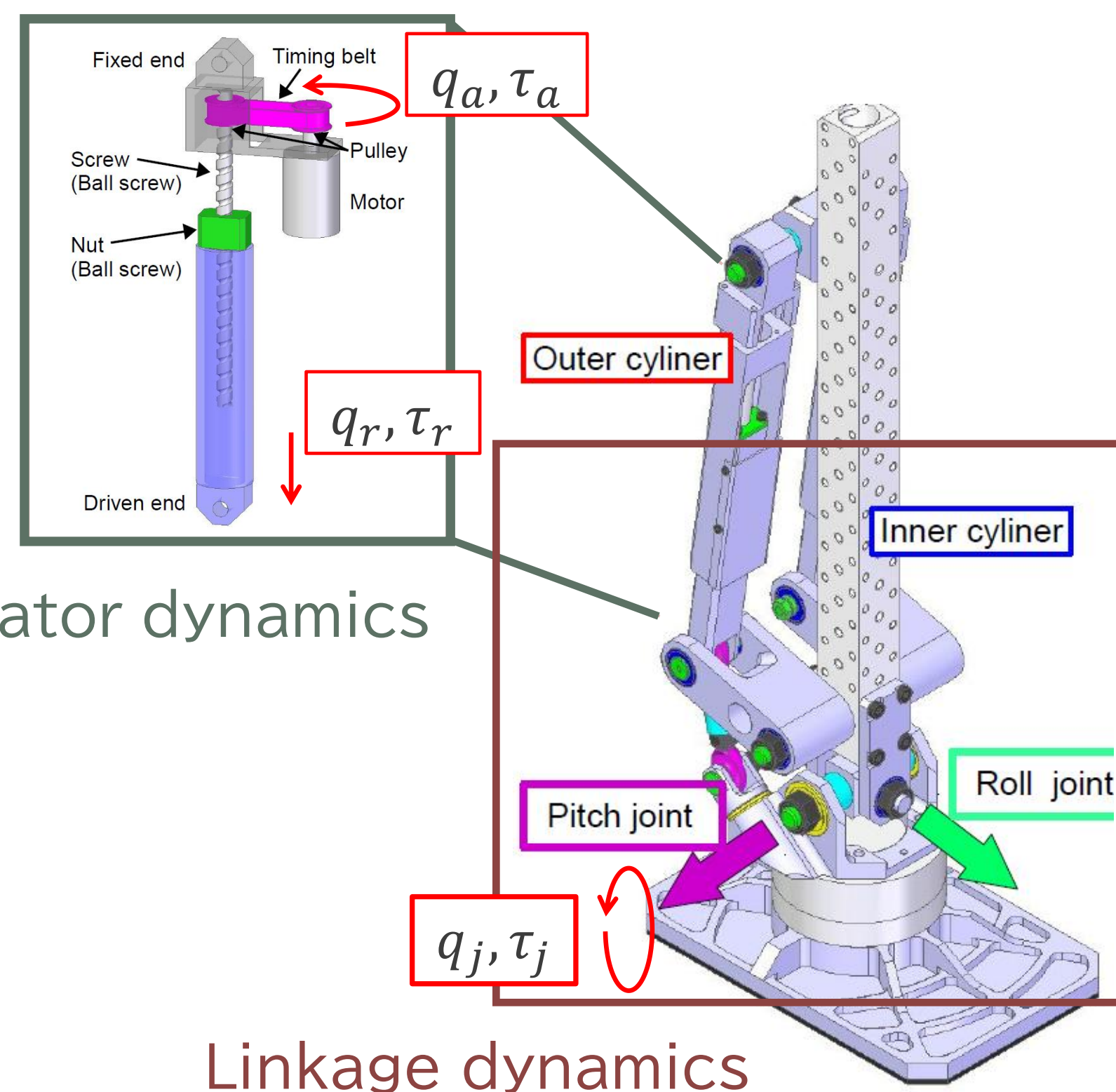
■ Objectives

Quantify the effects of linkage mechanisms on joint-level dynamics using an analytical model through system identification experiments

Joint-space Dynamics of Linkage-Driven System

■ Divide into two sub system

■ Kinematics



Actuator dynamics

Linkage dynamics

$$q_a = f_2(q_r)$$

$$q_r = f_1(q_j)$$

$$J_1 = \frac{\partial f_1}{\partial q_j} = \frac{\partial q_r}{\partial q_j} \quad J_2 = \frac{\partial f_2}{\partial q_r} = \frac{\partial q_a}{\partial q_r} \quad J = J_1 J_2$$

$$\dot{q}_r = J_1 \dot{q}_j \quad \dot{q}_a = J_2 \dot{q}_r = J \dot{q}_j$$

$$\ddot{q}_r = J_1 \ddot{q}_j + \dot{J}_1 \dot{q}_j \quad \ddot{q}_a = J \ddot{q}_j + \dot{J} \dot{q}_j$$

■ Dynamics

$$\tau_j = J^T (\tau_a - \tau_{f,a}(J\dot{q}_j)) - (J^T I_a J + J_1^T I_r J_1) \ddot{q}_j - (J^T I_a \dot{J} + J_1^T I_r \dot{J}_1) \dot{q}_j - J_1^T \tau_{f,r}(J_1 \dot{q}_j) - \tau_{f,j}(\dot{q}_j)$$

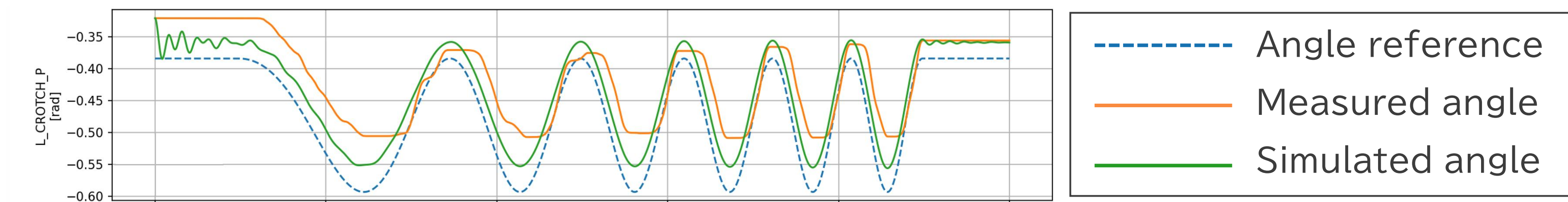
$$\tau_j = \underbrace{J^T \tau_a}_{\text{Torque transmission}} - \underbrace{M_{eq}(q_j) \ddot{q}_j}_{\text{Inertial}} - \underbrace{h_{eq}(q_j, \dot{q}_j)}_{\text{Velocity-dependent}} - \underbrace{\tau_{f,eq}(q_j, \dot{q}_j)}_{\text{Friction}}$$

Evaluation

■ System Identification (sys-ID) Experiment

- Compared position reconstruction accuracy across models
- Optimized model parameters to match real and simulated responses

Target joints: left-leg crotch, knee, and ankle pitch



■ Compared Models

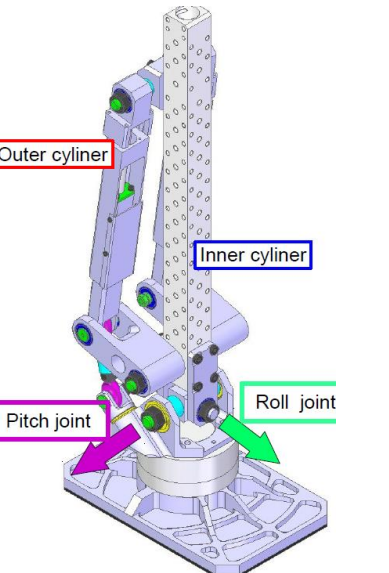
$$M_1: \text{Serial chain} \quad \tau_j = \underbrace{K_p(q_j^{ref} - q_j)}_{\text{Joint PD with constant gains}} - \underbrace{K_d \dot{q}_j}_{\text{Joint viscous + Coulomb}} - \underbrace{I_{const} \ddot{q}_j}_{\text{A constant inertia}} - F_c \text{sign}(\dot{q}_j) - D \dot{q}_j$$

M_2 : State-dependent torque mapping

$$\tau_j = J_1^T(q_j) \{ K_{p,r}(q_r^{ref} - q_r) - K_{d,r} \dot{q}_r \} - I_{const} \ddot{q}_j - F_c \text{sign}(\dot{q}_j) - D \dot{q}_j$$

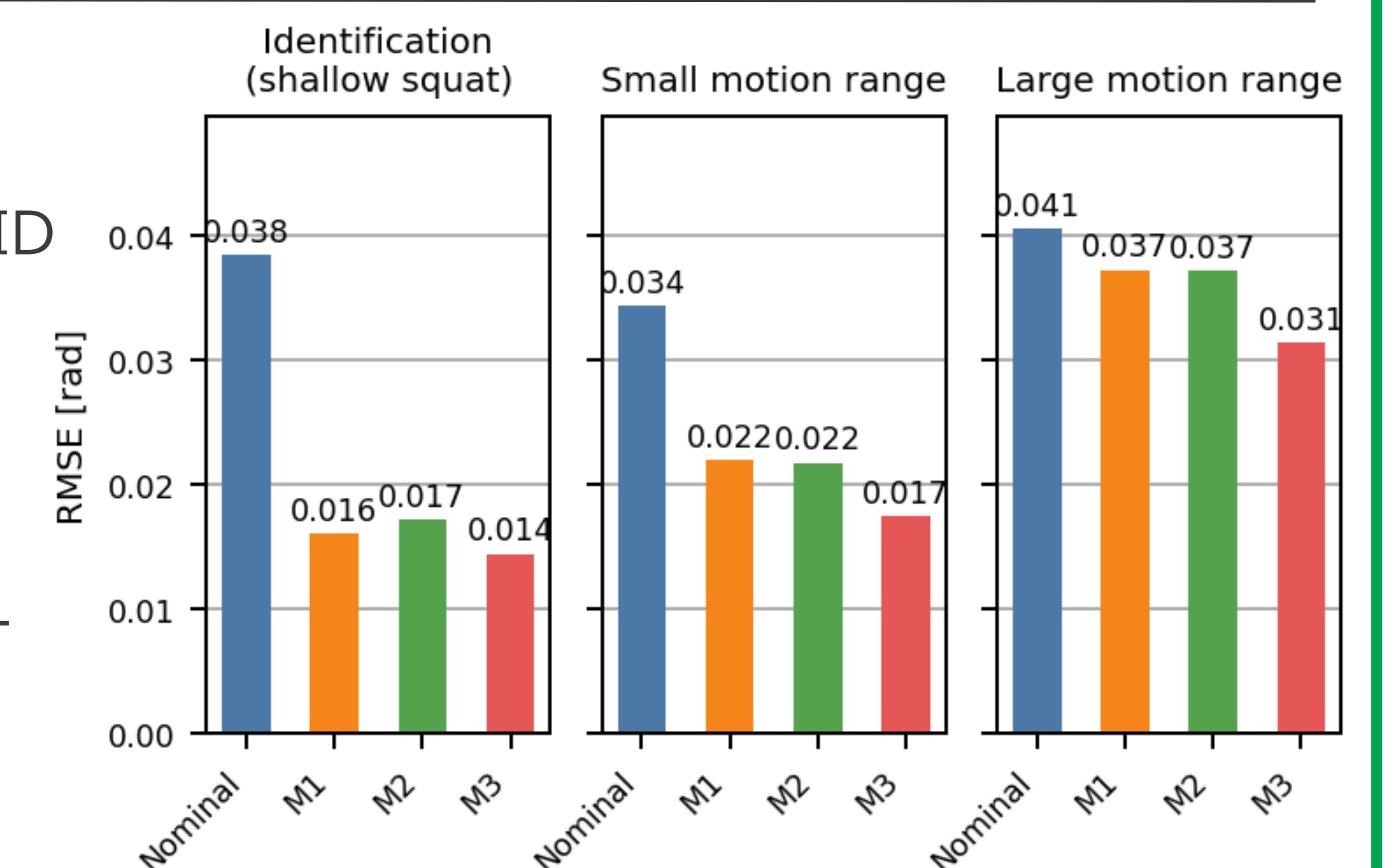
M_3 : Full linkage dynamics

$$\tau_j = J_1^T(q_j) \{ K_{p,r}(q_r^{ref} - q_r) - K_{d,r} \dot{q}_r \} - \underbrace{M_{eq}(q_j, I_r) \ddot{q}_j}_{\text{Rod PD with the actuator Jacobian}} - h_{eq}(q_j, \dot{q}_j) - J_1^T(q_j) \{ F_{c,r} \text{sign}(J_1 \dot{q}_j) + D_r J_1 \dot{q}_j \} - F_c \text{sign}(\dot{q}_j) - D \dot{q}_j$$



Results

- Similar RMSEs across models on the identification data, confirming successful sys-ID
- Larger errors for M1 and M2 on motions different from the identification motion
- These results suggest that detailed modeling, such as M3, becomes more important for large-range motions



Conclusion and Future Works

- Derived a joint-space dynamics model that explicitly consider the leg linkage
- The impact of linkage modeling may depend on the motion range used for learning
- Future work: Evaluate how this modeling strategy affects policy training

References

[1] Unitree Robotics, "Unitree mujoco," <https://github.com/unitreerobotics/unitree-mujoco/tree/main/unitree-robots>, 2026-07-14.
[2] F. Amadio et al., "Learning to walk with hybrid serial-parallel link ages: a case study on the kangaroo robot," in In 13th Int. Conf. on Robot Intelligence Technology and Applications, 2025.

Acknowledgements

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